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Beyond Accuracy: Building and Evaluating a 30-Day Readmission Model for Diabetic Patients

Introduction

Unexpected re-admissions are costly situations for hospitals, and many times patients too. They are also closely watched by regulators to ensure a facility's resources are being properly distributed and their care is making an impact on patients health. Diabetic patients, who tend to be a particularly high-risk, high-utilization group, can cycle through the hospital system within just weeks of discharge. (Tanner 2026). This is the kind of readmission frequency that sits on the fence between high costs attributed to genuine, consistent, required care for a struggling patient and ineffective procedures or standards that may be causing a them to return.

Based on 10 years of hospital admission data on patients with diabetes, from the UCI machine learning repository (Clore, Deshazo, Strack 2014), this project will build a predictive model or set of models to determine which diabetic inpatients are most likely to be readmitted within 30 days. After long lengths of stay a hospitals has limited follow-up resources that could be directed to a patient who needs them most. This often means redirecting follow-up resources away from a lower-risk patient and toward one the mode flags as high-risk, who need care now (Edwards, Thompson 2024). The upfront investment will pay off if the quality of that investment is enough to prevent another readmission later.

The consequences of doing this might lead to an overall increase in the quality of care, but also a greater avoidance of the federal penalties tied to consistent patient readmission.

Model Choices and Reasoning

Logistic regression is a strong baseline benchmark and is the clinical-risk-modeling standard (Christodoulou 2019). It can translate its coefficient into odds ratios that can be communicated to stakeholders, which makes it's findings easy to explain. It also is often compared and contrasted to linear regression, its numerical counterpart. However, it assumes the relationship between each predictor and readmission risk is linear on the log-odds scale. As a result, patterns and common interactions in medical data may go undetected because the model is constrained to that linear, additive form.

Random forest on the other hand, addresses that limitation by learning patterns directly form the data. It builds many decision trees, then combines and learns from them. This lets it capture non-linearities and interactions automatically and typically performs well on mess, high-dimensional data. The trade off with Random Forest is interpretability, because random forest does not produce clean coefficients of odds rations. Explaining an individual prediction requires additional interpretation techniques such as feature importance, which ranks how much each factor influenced the model, or SHAP values, which break down a single prediction to show how each factor pushed the risk up or down.

The final model is gradient boosting. On tabular clinical data, boosted trees usually win on raw predictive power. This is because they build trees sequentially, with each new tree trained specifically to correct the errors left by the ones before it, so the model keeps refining its weak spots until it captures subtle patterns a single pass would miss.

These three models will be bench-marked and compared with one another to see how the performance changes as the models move from a simple baseline to the more flexible learners. The best one will be recommended for use in the predictions along with an explanation as to why it was chosen.

Prospective Evaluation Metrics

Due to imbalance in the data, accuracy as an evaluation metric should be avoided. Only when data contains a 50/50 split will accuracy make sense. Instead the three main tiers of evaluation metrics will be divided into discrimination, calibration and fairness. Checking for discrimination means evaluating when a model makes an unfair or unequal decisions against specific groups of people. Calibration refers to the process of adjusting the predicted probabilities of a model, so they reflect the true likelihood of an event. Fairness ensures models do not unfairly treat people based on sensitive traits (like race or gender).

In the discrimination tier, metrics such as ROC-AUC and PR-AUC will be used to measure how well the model ranks patients by risk. ROC-AUC is especially best fit for this project because is is the standard in clinical work. For these metrics a threshold will be chosen to mark where a patient would actually be marked as “high-risk” in order to have a way to accurately report sensitivity. In the calibration tier, patients will be binned by predicted risk and plotted by the mean predicted probability against the observed readmission rate in each bin. As a reference, a perfectly calibrated model falls on a 45-degree line. A calibration slope will then be computed by regressing the observed outcomes on the model's predicted log-odds. A slope of 1 indicates perfect calibration, while a slope below 1 indicates predictions are too extreme, a common sign of overfitting, which is a particular risk with the gradient boosting model. To consider fairness, the discrimination and calibration metrics described above will be recomputed separately within demographic subgroups defined by race, age, and gender, so performance can be compared across groups. The primary fairness metric will be equal opportunity, which checks whether the model achieves similar sensitivity (true positive rate) across each group. In

other words, this metric is testing whether real readmissions are caught equally well regardless of a patient's demographic group.

Hopeful Takeaways

After having ran the models and evaluated them, sometimes even iterating between the those steps, a model should be completed that can accurately predict the readmission of a diabetes patient in the next 30 days, after a 14 day stay at most. Along the way, we will learn the strengths and weakness of how different models and evaluation metrics are used. It is expected that moving from a logic based approach (logistic regression) to a more pattern based learner (random forest and gradient boosting) will improve model accuracy.

Risks and Ethical Implications

Handling healthcare data is always sensitive because the data itself, based on the sector, (health insurance, hospitals, doctors offices etc.), can have different impacts on a patient if it is misused. In the industry this is called PHI or Personal Health Information. It can come in the form of actual health records, insurance claims or upon trigger of an insurance claim denial (will indicate possible emotional or financial distress). That being said, in the description of the dataset it is noted that there does contain data that might be considered sensitive. It contains information about age, gender and race of the patients. Thankfully though it does not include personally identifiable information, but those who created the dataset would have needed to be careful to separate the identifying data from the sensitive data that existed.

A potential risk might be that a patient with an unusual number of hospital vists is contributing to the model in a way that cause the metrics to be inaccurate. The model would then only be accurate in testing, but when deployed in the real world will likely performs poorly. So avoid this, outliers will be detected and data will be imputed, as it is also noted that there are missing values in the dataset.

Contingency Plan

In the event that a diabetic patient cannot be predicted to be readmitted within 30 days of discharge, the project will pivot to predicting length of stay based on patient condition and prior visit history. Because length of stay is already available in the same dataset, this shift would require changing only the target variable and evaluation metrics, not the data source.

Additional Notes

This project will attempt to predict the readmission of diabetes patients within 30 days of discharge after a 14 day stay at most. It is a binary classification problem that will be using logistic regression, random forest and gradient boosting as the predictive models. Then three different types of evaluation metrics will be applied to satisfy discrimination, calibration and fairness.

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